

Lecture 5

In-Lecture Questions

CS4243

A 224x224x3 image is processed by either (a) a fully connected layer with 64 outputs, or (b) a convolutional layer with 64 filters of size 3x3. If the image resolution doubles to 448x448, what happens to the number of learnable weights?



The fully connected layer has four times as many; the convolutional layer is unchanged.

82%

The fully connected layer doubles; the convolutional layer quadruples.

10%

Both layers have four times as many weights

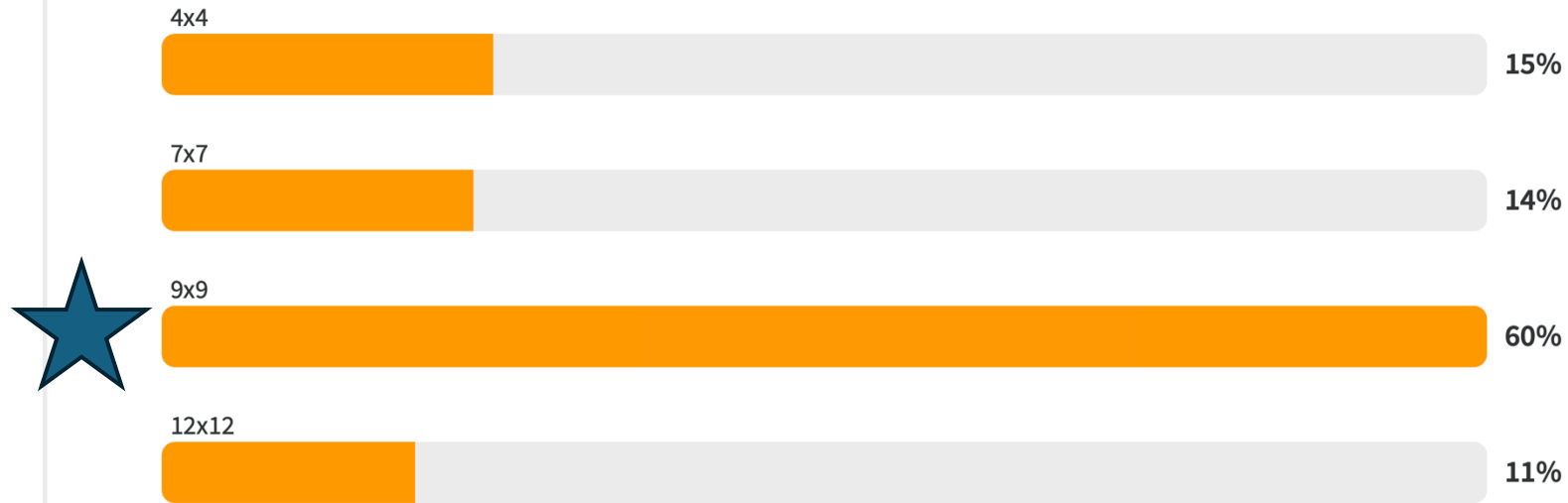
8%

Both layers have the same number of weights as before.

0%

- The fully connected layer has one weight per input–output connection, so doubling both spatial dimensions quadruples its parameters.
- The convolution still has the same number of weights because the same filters are shared across locations. Its computation and activation-map size increase, but its parameter count does not.

Consider four consecutive 3x3 convolutional layers, each with stride 1 and no dilation and padding of 1. What is the receptive field of an activation in the fourth layer?



- $R = 1 + 4(3 - 1) = 9$.
- The visualization on the right shows 2 convolution layers; abstract to 4 layers we will have 9x9

